

Literature status: slicing and thin-shell conjectures

Run `fa_banach_001`; source arXiv:1310.1204

Source conjectures

Guedon's survey *Concentration phenomena in high dimensional geometry* records several major conjectures in high-dimensional convex geometry. This status note concerns two named conjecture families from the source:

- Conjecture **slicing**, the hyperplane conjecture/Bourgain slicing problem: for every volume-one convex body $K \subset \mathbb{R}^n$ with barycenter at the origin, some hyperplane section $K \cap \theta^\perp$ has $(n-1)$ -dimensional volume bounded below by a universal positive constant.
- Conjectures **ThinShell** and **ThinShellbis**: for every isotropic log-concave random vector $X \in \mathbb{R}^n$, respectively

$$\mathrm{Var} |X|_2^2 \leq C \mathbb{E} |X|_2^2 \quad \text{and} \quad \mathrm{Var} |X|_2 \leq C$$

with a universal constant C .

Supporting literature

Klartag and Lehec's paper *Affirmative Resolution of Bourgain's Slicing Problem using Guan's Bound* explicitly states that it gives the final step in the affirmative resolution of Bourgain's slicing problem. Its main theorem asserts that for any convex body $K \subseteq \mathbb{R}^n$ of volume one, there is a hyperplane H such that

$$\mathrm{Vol}_{n-1}(K \cap H) > c$$

for a universal constant $c > 0$. This is the source's Conjecture **slicing**.

Klartag and Lehec's later paper *Thin-shell bounds via parallel coupling* states the following theorem.

Theorem 1 (Klartag–Lehec, arXiv:2507.15495, Theorem 1.1). *Let X be an isotropic log-concave random vector in $\mathbb{R}^n$. Then*

$$\mathrm{Var}(|X|^2) = \mathbb{E}(|X|^2 - n)^2 \leq Cn,$$

where $C > 0$ is a universal constant.

Since isotropy gives $\mathbb{E}|X|^2 = n$, this is exactly Guedon's Conjecture **ThinShell**. Moreover,

$$\mathrm{Var} |X| \leq \mathbb{E}(|X| - \mathbb{E}|X|)^2 \leq \mathbb{E}(|X| - \sqrt{n})^2 \lesssim 1,$$

where the last bound follows from the same thin-shell theorem as stated in the paper's abstract. Thus Conjecture **ThinShellbis** is also answered.

Status

The source’s slicing conjecture and variance thin-shell conjectures are already answered by later literature:

`slicing` by arXiv:2412.15044,

and

`ThinShell, ThinShellbis` by arXiv:2507.15495.

This packet is a literature-status identification and contains no new proof from the run.

Limitations

This packet does not claim that all conjectures in arXiv:1310.1204 are solved. In particular, it does not record the Kannan–Lovasz–Simonovits conjecture as solved, does not claim the stronger all- t exponential-tail concentration inequality from Conjecture **Concentration**, and does not address Latala’s weak/strong moments conjecture.

Verification

Before packaging, the run lightweight indexes were searched for arXiv:1310.1204, the source title, “slicing”, “thin shell”, “KLS”, “concentration phenomena”, and weak/strong moment terminology. No existing exact packet for this source was found. Web and arXiv checks on 2026-06-16 identified arXiv:2412.15044 and arXiv:2507.15495 as the decisive supporting papers.

References

- [1] O. Guedon, *Concentration phenomena in high dimensional geometry*, arXiv:1310.1204.
- [2] B. Klartag and J. Lehec, *Affirmative Resolution of Bourgain’s Slicing Problem using Guan’s Bound*, arXiv:2412.15044.
- [3] B. Klartag and J. Lehec, *Thin-shell bounds via parallel coupling*, arXiv:2507.15495.